

Lecture - 39

Solution to $\dot{\vec{x}} = A\vec{x}$ when A has Complex eigenvalues:-

$A \in \mathbb{R}^{n \times n} \rightarrow$ if Complex eigenvalues occur

$\downarrow$
Complex Conjugate pairs

$\downarrow$
if $\vec{v}_1$ is an eigenvector associated with Complex eigenval λ_1 , then $\vec{v}_1^*$ is an eigvec associated with eigenval λ_1^*

Example:-

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad \text{eig.val.} \rightarrow \lambda_1 = i \quad \text{and} \quad \lambda_2 = -i$$

$(P+iQ) [P=0, Q=1] \quad (P-iQ)$

$\downarrow$

$$\text{eigvec} \quad \vec{v}_1 = \begin{bmatrix} 1 \\ -i \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

Let $\vec{v}_1 = \vec{a} + i\vec{b}$ and $\vec{v}_2 = \vec{a} - i\vec{b} \rightarrow$ are $\vec{a}$ & $\vec{b}$ eigen vectors? $\rightarrow \underline{\underline{\text{No}}}$

for this example $\vec{a} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$

$$\begin{aligned} A\vec{a} &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad A\vec{b} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \\ &= -\vec{b} \\ &= -\phi \vec{b} \end{aligned} \quad \begin{aligned} & \\ & \underbrace{\begin{bmatrix} 0 \\ -1 \end{bmatrix}}_{\vec{a}} \\ &= \phi \vec{a} \end{aligned}$$

for constructing the fundamental set:-

$$\lambda_1 = e^{(P+jQ)t} \quad \vec{u}_1 = \vec{a} + j\vec{b}$$
$$\lambda_2 = e^{(P-jQ)t} \quad \vec{u}_2 = \vec{a} - j\vec{b}$$

$$\begin{aligned}\vec{y} &= (a+jb)e^{(P+jQ)t} \\ &= e^{Pt} e^{jQt} (\vec{a} + j\vec{b}) \\ &= e^{Pt} [\cos Qt + j \sin Qt] (\vec{a} + j\vec{b}) \\ &= e^{Pt} [(\vec{a} \cos Qt - \vec{b} \sin Qt) + j(\vec{b} \cos Qt + \vec{a} \sin Qt)]\end{aligned}$$

check if $\vec{x}_1 = e^{Pt} [\vec{a} \cos Qt - \vec{b} \sin Qt]$ and $\vec{x}_2 = e^{Pt} [\vec{b} \cos Qt + \vec{a} \sin Qt]$
can be solutions?

check for $\vec{x}_1$:- $\dot{\vec{x}}_1 = e^{Pt} \{ [P\vec{a} - Q\vec{b}] \cos Qt - [P\vec{b} + Q\vec{a}] \sin Qt \} \rightarrow (1)$

$$A\vec{x}_1 = e^{Pt} [A\vec{a} \cos Qt - A\vec{b} \sin Qt] \rightarrow (2)$$

for $\vec{x}_1$ to be a solution $(1) = (2)$, which requires $\left. \begin{aligned} A\vec{a} &= P\vec{a} - Q\vec{b} \\ A\vec{b} &= P\vec{b} + Q\vec{a} \end{aligned} \right\} \begin{array}{l} \text{True, result from} \\ \text{Linear alg.} \end{array}$

So $\vec{x}_1$ is a sol and so is $\vec{x}_2$ (can be shown)

also $\vec{x}_1$ & $\vec{x}_2$ form a fundamental set.

for $A \in \mathbb{R}^{2 \times 2}$ with Complex Conjugate eig. val:

$$\vec{x}_1 = e^{Pt} [\vec{a} \ \vec{b}] \begin{bmatrix} \cos Qt \\ -\sin Qt \end{bmatrix} \text{ and } \vec{x}_2 = e^{Pt} [\vec{a} \ \vec{b}] \begin{bmatrix} \sin Qt \\ \cos Qt \end{bmatrix}$$

Sol. matrix $X = [\vec{x}_1 \ \vec{x}_2] = e^{Pt} [\vec{a} \ \vec{b}] \underbrace{\begin{bmatrix} \cos Qt & \sin Qt \\ -\sin Qt & \cos Qt \end{bmatrix}}_{R_\Theta}$

if $P > 0 \rightarrow$ spiral to ∞
 $P < 0 \rightarrow$ " to 0

if $P = 0 \rightarrow$ Trajectory $\rightarrow$ Spiral or a Circle.

So $X = e^{Pt} [\vec{a} \ \vec{b}] R_\Theta$

for the example:-

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \text{ eig. val. } \rightarrow \lambda_1 = i \text{ and } \lambda_2 = -i$$

$(P+iQ) [P=0, Q=1] \downarrow (P-iQ)$
 eig vec $\vec{v}_1 = \begin{bmatrix} 1 \\ -i \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ i \end{bmatrix}$

$$\vec{a} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$X = e^{Pt} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$$

$$X = e^{Pt} \begin{bmatrix} \cos t & \sin t \\ \sin t & -\cos t \end{bmatrix} \leftarrow \text{Trajectory ??}$$